

In just a few years, Generative Artificial Intelligence (Gen-AI) has gone from a tool of limited capacity, only used by a few early-adopters to a powerful and inescapable resource embedded in our daily lives. Over time, research suggests that Gen-AI could impact the future of work. For example, in some fields, Gen-AI may replace humans (or heavily reduce the human workload), while other fields might not be heavily impacted or might even grow. In this question, you will explore how post-secondary educational institutions, of various types, should best prepare their future graduates in light of this new technology. Specifically, you are asked to do the following.

- Choose three careers, one from each of the following categories:
 - STEM career: people in this career often have at least a four year university degree in the sciences, engineering, or mathematics;
 - Trade career: people in this career often have training from a trade school and/or an apprenticeship program, such as chef, plumber, and electrician;
 - Arts career: people in this career often have studied at an arts school, conservatory, or cultural center, such as musician, dancer, or painter.
- Design a data-informed model to explore the future of each of your three chosen professions, given the current trajectory and expected impacts of Gen-AI. Be sure to identify your data sources as well as your reasoning behind any drivers that you expect to change this profession as a result of Gen-AI. Note: you may leverage existing research on the future of work, but be sure to cite your sources and explain how you are using the established research to inform your analysis.
- Identify a specific post-secondary institution and program of study for each career you are analyzing (one at a university, one at a trade school, one at an arts school), and focus your recommendations accordingly. In other words, you should have three sets of recommendations that address the following question: Based on your analysis, how would you advise the leaders of each of these institutions to address Gen-AI in the programs specific to the careers you are analyzing? Below are just some thoughts you may wish to consider; teams should not attempt to address all of these ideas, but should use these as inspiration that will lead to a cogent and thorough analysis that should vary team to team. | ©2026 by COMAP | www.comap.org | www.mathmodels.org | info@comap.org |
- Should the program of study grow or shrink (graduate more or fewer people) as a result of changes in the career due to Gen-AI? If the field should grow, how might the institution recruit more people; and if the field should shrink, are there other programs in the school that should grow to absorb the people who used to study in this program?
- What should these three different programs of study teach about Gen-AI? Many post-secondary institutions of learning have asked this question and are still developing their response. While some institutions have outright banned

discussions over detection-based policies, and balance technical AI literacy with domain-specific competencies that AI cannot replace—an approach that demonstrates measurable improvements in both student learning outcomes and graduate employability readiness.

Abstract

Generative AI is transforming skill requirements across career categories, with STEM fields experiencing the most documented impact through tools for code generation, debugging, and automated testing, while trade careers (hospitality, aviation) and arts careers (interior design, creative media) receive significantly less research attention. Evidence suggests GenAI is unlikely to replace graduates in high cognitive skill jobs but will enhance efficiency and transform job responsibilities, with industry partners emphasizing critical thinking (22%), problem-solving (18%), and adaptability (12%) as essential competencies. Structured pedagogical approaches demonstrate measurable effectiveness: the SPIRAL framework (Skills Practiced Independently, Revisited with AI Later) produces positive changes in both domain-specific and GenAI self-efficacy, while the AI-Lab intervention generates large effect sizes in student comfort and mindful AI usage. Students who develop foundational skills before AI integration show better critical evaluation capabilities, with 84% of groups in structured programs moving beyond passive acceptance of AI suggestions.

Institutional responses remain inconsistent—93% of surveyed medical schools lack GenAI policies, while Academic Integrity Officers prioritize assessment redesign (88.2%) and ethical discussions (76.5%) over detection tools (39.5%). A significant academic-industry gap exists, with GenAI competency increasingly valued in industry but underemphasized in higher education, and only 54% of students believing employers will expect such skills. For STEM programs, validated frameworks combining longitudinal AI literacy scaffolding with validation task requirements show promise. For trade education, performance expectancy and facilitating conditions emerge as key adoption determinants, requiring infrastructure support before curriculum integration. For arts education, the “demystification” effect—where exposure transitions students from fear of AI displacement to nuanced understanding—suggests that structured engagement can enhance rather than diminish creative development. The three-theme framework from medical education—learning about GenAI, learning with GenAI, and learning aside from GenAI to preserve competencies AI cannot replace—offers a generalizable model across institution types.

Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: “In just a few years, Generative Artificial Intelligence (Gen-AI) has gone from a tool of limited capacity, only used by a few early-adopters to a powerful and inescapable resource embedded in our daily lives. Over time, research suggests that Gen-AI could impact the future of work. For example, in some fields, Gen-AI may replace humans (or heavily reduce the human workload), while other fields might not be heavily impacted or might even grow. In this question, you will explore how post-secondary educational institutions, of various types, should best prepare their future graduates in light of this new technology. Specifically, you are asking do to the following. ● Choose three careers, one from each of the following categories: ○ STEM career: people in this career often have at least a four year university degree in the sciences, engineering, or mathematics; ○ Trade career: people in this career often have training from a trade school and/or an apprenticeship program, such as chef, plumber, and electrician; ○ Arts career: people in this career often have studied at an arts school, conservatory, or cultural center, such as musician, dancer, or painter. ● Design a data-informed model to explore the future of each of your three chosen professions, given the current trajectory and expected impacts of Gen-AI. Be sure to identify your data sources as

well as your reasoning behind any drivers that you expect to change this profession as a result of Gen-AI. Note: you may leverage existing research on the future of work, but be sure to cite your sources and explain how you are using the established research to inform your analysis. ● Identify a specific post-secondary institution and program of study for each career you are analyzing (one at a university, one at a trade school, one at an arts school), and focus your recommendations accordingly. In other words, you should have three sets of recommendations that address the following question: Based on your analysis, how would you advise the leaders of each of these institutions to address Gen-AI in the programs specific to the careers you are analyzing? Below are just some thoughts you may wish to consider; teams should not attempt to address all of these ideas, but should use these as inspiration that will lead to a cogent and thorough analysis that should vary team to team. | ©2026 by COMAP | www.comap.org | www.mathmodels.org | info@comap.org | ○ Should the program of study grow or shrink (graduate more or fewer people) as a result of changes in the career due to Gen-AI? If the field should grow, how might the institution recruit more people; and if the field should shrink, are there other programs in the school that should grow to absorb the people who used to study in this program? ○ What should these three different programs of study teach about Gen-AI? Many post-secondary institutions of learning have asked this question and are still developing their response. While some institutions have outright banned the use of AI on any assignments, others have brought the use of AI to the forefront of their curriculum. Some schools aim to produce experts who can contribute to the leading edge of the technological field, while some focus on graduating students in non-technical fields who are fluent users of the technology. Some institutions encourage their students to think about all the ways they can apply this new technology, and some schools challenge students to carefully weigh the benefits and costs of using AI, given the requisite energy usage, water demands, and risk of insufficient (often missing or incorrect) attribution to the original creators of ideas or content. For the three programs of study at the three institutions you've selected, what do you recommend to best support the employability of their graduates? Be sure to support your recommendations with the results of a mathematical model. ○ While this problem poses the question through the context of employability of graduates in a world where Gen-AI is ubiquitous, perhaps employment demands are not the only way to measure the success of the institutional policies you are proposing. What other factors do you believe should be considered, and how do your models and recommendations change when you consider these other factors? ○ If you believe your specific recommendations can be generalized beyond one institution and/or beyond one program, be sure to explain the extent of the generalization and justify this. Task: Deeply analyze the competition problem (see PDF file). Identify the core models that may be used behind the problem."

The search returned 1,000 total results from Elicit.

We retrieved 1000 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Post-Secondary Education Setting:** Does this study examine universities, colleges, trade schools, vocational institutions, arts schools, conservatories, or other post-secondary educational institutions and their programs (not K-12 education or corporate training programs)?
- **Generative AI Focus:** Does this study investigate educational approaches, curriculum modifications, teaching methods, or institutional policies specifically related to generative artificial intelligence tools (e.g., ChatGPT, DALL-E, GitHub Copilot, etc.)?
- **Career Preparation and Future of Work:** Does this study address how educational programs prepare students for future careers, employment outcomes, workforce readiness, OR how generative AI impacts career prospects and job market changes?

- **Relevant Career Fields:** Does this study examine programs or careers in science, technology, engineering, mathematics, skilled trades, or creative arts fields?
- **Empirical Research Design:** Is this study primary empirical research (quantitative, qualitative, or mixed methods), a systematic review, meta-analysis, or rigorous case study with clear methodology (not an opinion piece, editorial, or commentary without empirical data)?
- **Measurable Outcomes:** Does this study include measurable outcomes related to educational effectiveness, career preparation, employment rates, skill assessments, curriculum effectiveness, student performance, or institutional adaptation metrics?
- **Educational Context Focus:** Does this study focus on educational preparation or institutional responses to generative AI (not solely on workplace AI implementation without educational components)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Career Field Analysis:**

Extract details about the specific careers, professions, or academic fields studied in relation to GenAI impact, including:

- Specific career/profession name
- Whether it falls into STEM, Trade, Arts, or other categories
- Current employment characteristics (education requirements, typical roles)
- Specific GenAI tools or applications relevant to this field

- **GenAI Impact Assessment:**

Extract evidence about how GenAI is currently impacting or projected to impact the studied careers/fields, including:

- Job displacement risks or job creation potential
- Changes in skill requirements or job responsibilities
- Productivity or efficiency changes
- New opportunities or threats identified
- Timeline projections for impacts
- Data sources or models used for impact assessment

- **Institutional Strategies:**

Extract specific strategies, policies, or approaches that educational institutions (universities, trade schools, arts schools) are implementing or recommending to address GenAI, including:

- Curriculum integration approaches
- Policy changes (bans, requirements, guidelines)
- Faculty development initiatives
- Infrastructure or technology investments
- Industry partnership strategies
- Program expansion or reduction decisions

- **Skills and Competencies:**

Extract information about new or changing skills and competencies identified as necessary for graduates in the GenAI era, including:

- Technical GenAI skills (prompt engineering, tool usage, etc.)
- Critical thinking and evaluation skills for AI outputs
- Ethical awareness and responsible AI use
- Human-AI collaboration abilities
- Domain-specific skills that complement AI
- Lifelong learning and adaptability skills

- **Pedagogical Changes:**

Extract details about how teaching methods and assessment approaches are changing to incorporate GenAI, including:

- Traditional vs. innovative assessment methods
- GenAI-incorporated learning activities
- Changes in assignment design
- Student evaluation methods
- Faculty teaching approaches
- Learning outcome modifications

- **Academic-Industry Gap:**

Extract evidence about gaps or alignment between academic GenAI preparation and workplace GenAI usage, including:

- Differences in how GenAI is used in academic vs. professional settings
- Industry feedback on graduate preparedness
- Workplace GenAI skill requirements not addressed in education
- Successful bridging strategies between academic and professional contexts

- **Implementation Evidence:**

Extract evidence about the effectiveness of institutional GenAI strategies and outcomes achieved, including:

- Measured improvements in student learning or engagement
- Graduate employment outcomes
- Industry satisfaction with graduate preparedness
- Student proficiency gains with GenAI tools
- Challenges or failures encountered
- Comparison data between different approaches

- **Future Models:**

Extract information about predictive models, frameworks, or projections for how careers and education should evolve with GenAI, including:

- Quantitative models or data-driven predictions
- Theoretical frameworks for GenAI integration
- Scenario planning or future projections

- Recommendations for program growth/reduction
- Generalizability claims across institutions or fields

Characteristics of Included Studies

The systematic review encompasses 40 sources examining GenAI's impact on higher education and career preparation across diverse fields and institutional contexts. The studies span multiple countries, methodologies, and educational levels.

Study	Full text retrieved?	Study Type	Career Field Focus	Geographic Context
M. S. Abdelaal et al., 2025	No	Primary survey study	STEM, Clinical/Health Sciences, Business, Education	Australia
Troy Schotter et al., 2025	Yes	Mixed methods longitudinal	Creative media, Game development, Filmmaking	United States
Vahid Garousi et al., 2025	Yes	Design-based research	Software Engineering	UK and Azerbaijan
Khinsa Fairuz Zahirah et al., 2025	Yes	Qualitative pilot study	Food Technology, Medicine, Hospitality, Psychology	Indonesia
Xiaojing Weng et al., 2024	Yes	Scoping review	Programming, Writing	International
Yoram Reich et al., 2025	Yes	Case study with survey	Systems Engineering	Israel
I. Shimizu et al., 2023	No	Qualitative workshop study	Medical/Physicians	Japan
Jonathan W. Browning et al., 2024	No	Qualitative evaluation	Data Science	United Kingdom
J. A. Ovi et al., 2025	Yes	Cross-sectional survey	Computer Science, Electrical Engineering	United States
N. Roy et al., 2025	Yes	Benchmarking study	Software Engineering	International
Tony Clear et al., 2025	Yes	Mixed methods (review + interviews)	IT Professionals	New Zealand and Sweden
Armin Alimardani et al., 2024	Yes	Empirical comparative study	Legal profession	Australia
Mahmoud Elkhodr et al., 2025	No	Case study	Systems Analysis	Australia

Study	Full text retrieved?	Study Type	Career Field Focus	Geographic Context
Mahmoud Elkhodr et al., 2025a	Yes	Case study	Cybersecurity Management	Australia
Meghan Reading Turchioe et al., 2024	No	Intervention study	Nursing	United States
Remco Jongkind et al., 2025	Yes	Cross-sectional survey	Medical Informatics	Netherlands
Anya Bardach et al., 2025	No	Mixed methods evaluation	Computer Science	United States
Cari Hyde-Vaamonde et al., 2025	Yes	Three-stage intervention	Legal professionals	United Kingdom
Milena Nikolic et al., 2025	No	NLP and ML analysis	Computer Science/Programming	Serbia
A. Tuzmen et al., 2024	No	Innovative practice study	Software Engineering	Not specified
Clara Rispler et al., 2025	Yes	Survey study (TAM framework)	Information Systems, Economics	Israel
Ahmad Tajudin Baharin et al., 2025	Yes	Quantitative survey (UTAUT)	Engineering, Hospitality, Business	Malaysia
Yume Menghe Xu et al. (n.d.)	No	Pre-post survey study	Engineering	United States
Ethan Dickey et al., 2025	Yes	Mixed methods intervention	Computer Science	United States
Andrea L. Irish et al., 2025	Yes	Qualitative descriptive	Software Development, Aviation, Arts	Not specified
T. Chiu et al., 2024	No	Qualitative thematic analysis	General higher education	International
Title Pending 6645	No	Mixed methods case study	Engineering	Not specified
Tsunagu Ichikawa et al., 2025	No	Descriptive cross-sectional survey	Medical (Osteopathic)	United States
Kate Tran et al., 2024	Yes	Framework development	General higher education	Australia
Silvia Albano et al., 2025	No	Practice-based methodology	Interior Design	Not specified
M. Shen et al., 2025	Yes	Qualitative study	Nursing	Not specified
John Robert Bautista et al., 2025	No	Descriptive qualitative	Nursing	Not specified
Anatolij Prohorov et al., 2024	No	Survey study	General (unspecified)	Ukraine and Latvia
Hauke Sandhaus et al., 2025	No	Qualitative interview study	Human-Computer Interaction	Not specified

Study	Full text retrieved?	Study Type	Career Field Focus	Geographic Context
Amy N Adkins et al. (n.d.)	Yes	Pre-post survey study	Biomedical Engineering	United States
Mostafa Mohamad et al., 2025	Yes	Qualitative thematic analysis	Computer Engineering	Middle East and North Africa
Yasaman Mohammadi et al., 2024	Yes	Participatory design study	General higher education	Australia
Miaoxian Wu et al., 2025	No	Quantitative clustering analysis	Customer Relationship Management	China
Patrick Rupprecht et al., 2025	No	Multi-method research	General (tertiary and vocational)	Austria
B. M. Aucoin et al. (n.d.)	Yes	Work-in-progress study	Engineering Leadership	United States

The included studies demonstrate substantial geographic diversity, spanning North America, Europe, Asia, Australia, and the Middle East. The predominant focus is on STEM careers, particularly software engineering, computer science, and healthcare professions. Notably, trade careers and arts careers receive significantly less attention in the literature, with only a few studies examining hospitality, interior design, and creative media. This disparity highlights a significant gap in current research regarding GenAI's impact on non-university educational pathways.

Career Field Analysis by Category

STEM Careers

The majority of included studies focus on STEM careers, providing substantial evidence regarding GenAI's impact on these fields.

Career Field	Key GenAI Applications	Skill Changes Identified	Sources
Software Engineering	Code generation, automated testing, debugging support	Critical thinking, validation of AI outputs, professional judgment	6 studies
Computer Science	Coding, debugging, writing tasks	Prompt engineering, tool usage	5 studies
Medical/Nursing	Clinical decision support, care plan development	Clinical reasoning, ethical GenAI usage	5 studies
Data Science	Data analysis, decision-making	Analytical capabilities, prompt engineering	2 studies
Engineering (General)	Design challenges, programming assistance	Critical thinking, creativity	4 studies
Cybersecurity	Policy drafting, risk assessment	Regulatory compliance, analytical reasoning	1 study

Career Field	Key GenAI Applications	Skill Changes Identified	Sources
Legal Profession	Legal writing, document analysis	Critical legal analysis, AI literacy	2 studies

Software engineering emerges as the most extensively studied STEM field, with GenAI tools like GitHub Copilot transforming code generation, debugging, and testing workflows. Students in these programs are expected to develop skills in validating AI-generated outputs and maintaining critical thinking despite AI assistance.

Trade Careers

Evidence regarding trade careers is notably sparse in the literature.

Career Field	Key GenAI Applications	Skill Changes Identified	Sources
Hospitality/Tourism	Customer service automation, content creation	Problem-solving, digital skills	2 studies
Aviation	Email automation, document templates, reports	Efficiency enhancement	1 study

The Technical and Vocational Education and Training (TVET) sector shows growing interest in GenAI integration, with performance expectancy and facilitating conditions identified as significant determinants of student adoption. However, challenges include technological literacy gaps and ethical concerns about plagiarism and data privacy.

Arts Careers

Arts-related careers receive the least attention in current research.

Career Field	Key GenAI Applications	Skill Changes Identified	Sources
Interior Design	Visualization, material moodboards, spatial composition	Client-centered design, iterative refinement	1 study
Creative Media	Content generation, filmmaking support	Critical judgment, ethical AI use	1 study
General Arts	Content generation, brainstorming	Creative process adaptation	1 study

The interior design study using Midjourney demonstrates that GAI can significantly improve students' ability to visualize and refine design proposals, with iterative feedback enabling real-time insights into design decisions. Creative media students showed positive changes in self-efficacy and experienced "demystification," transitioning from fear about AI taking over jobs to more nuanced perspectives.

GenAI Impact Assessment

Job Displacement and Creation Dynamics

The literature presents a nuanced picture of GenAI's impact on employment across sectors.

Impact Type	Evidence	Career Context	Source
Job displacement risk	GenAI automates tasks in data entry and report drafting	Cross-sector	Andrea L. Irish et al., 2025
Job transformation	AI unlikely to replace graduates in high cognitive skill jobs but will enhance efficiency	Legal profession	Armin Alimardani et al., 2024
Skill erosion risk	Over-reliance could lead to skill fade and erosion of fundamental skills	Computer Engineering	Mostafa Mohamad et al., 2025
Reduced fear of displacement	Students transitioned from fear about AI taking jobs to doubting AI capability	Creative media	Troy Schotter et al., 2025
Polarized perceptions	Bimodal distribution of "P(doom)" estimates among engineering students	Engineering	J. A. Ovi et al., 2025

Industry partners consistently emphasize that critical thinking (22%), problem-solving (18%), and adaptability (12%) are increasingly important skills. This suggests that GenAI will transform rather than eliminate most careers, shifting emphasis from routine tasks to higher-order competencies.

Changes in Skill Requirements

Skill Category	Specific Skills	Evidence Source
Technical GenAI Skills	Prompt engineering, tool usage, iterative prompt refinement	Multiple studies
Critical Evaluation	Assessing AI outputs, identifying hallucinations, validating accuracy	Multiple studies
Ethical Awareness	Data privacy, algorithmic bias, intellectual property	Multiple studies
Human-AI Collaboration	Peer-review of AI artifacts, human-guided AI partnership	Multiple studies
Domain Expertise	Clinical judgment, legal analysis, design thinking	Field-specific studies

Students are expected to move beyond passive acceptance of AI suggestions, with 84% of groups in one study showing selective judgment when using GenAI tools. However, significant gaps remain: 55% of students struggled to identify when AI misclassified system boundaries, and 45% missed data management errors.

Institutional Strategies and Policies

Curriculum Integration Approaches

Institutions have adopted diverse strategies for integrating GenAI into curricula:

Strategy	Description	Implementation Example	Source
SPIRAL Integration	Skills Practiced Independently, Revisited with AI Later	Undergraduate creative media course	Troy Schotter et al., 2025
SAGE Framework	Structured AI-Guided Education for orchestration skills	Systems analysis across four universities	Mahmoud Elkhodr et al., 2025
Three-Tier Three-Driver Model	Personalized teaching based on career trajectories	Customer Relationship Management course	Miaoxian Wu et al., 2025
GenAI-CT Framework	GenAI-driven activities with critical thinking exercises	Undergraduate engineering course	Title Pending 6645
AI-Lab Intervention	Guided scaffolding and mindful engagement	Data Structures and Algorithms courses	Ethan Dickey et al., 2025
AI Immersion Program	Four-stage program covering concepts to projects	Multi-disciplinary undergraduate program	Khinsa Fairuz Zahirah et al., 2025

The SPIRAL approach demonstrates particular promise, with students developing domain skills without AI before integrating AI tools, leading to positive changes in both creative media self-efficacy and generative AI use self-efficacy.

Policy Responses

Institutional policy responses vary significantly:

Policy Type	Prevalence	Example	Source
No formal GenAI policy	93% of surveyed medical schools	US Osteopathic Medical Schools	Tsunagu Ichikawa et al., 2025
Assessment redesign prioritized	88.2% of Academic Integrity Officers	Australian university	M. S. Abdelaal et al., 2025
Detection tools deprioritized	Only 39.5% recommend	Australian university	M. S. Abdelaal et al., 2025

Policy Type	Prevalence	Example	Source
Ethical discussions emphasized	76.5% of Academic Integrity Officers	Australian university	M. S. Abdelaal et al., 2025
AI statement requirements	Required in syllabi	Purdue University	Ethan Dickey et al., 2025

Notably, while 90% of Academic Integrity Officers reported investigating GenAI-related misconduct, only 21% felt confident in their ability to detect it, highlighting the inadequacy of detection-focused approaches.

Faculty Development

Initiative Type	Description	Source
Faculty training programs	Strategies for responsibly allowing and assessing GenAI use	Vahid Garousi et al., 2025
AI-focused workshops	Professional development for AI integration	Andrea L. Irish et al., 2025
Mentorship models	Faculty guiding students through GenAI integration	Khinsa Fairuz Zahirah et al., 2025
Tool provision	Subscription to ChatGPT 4.0 for students	Meghan Reading Turchioe et al., 2024

Faculty development remains inconsistent across institutions, with many lacking formal training programs for GenAI integration.

Pedagogical Changes and Assessment Innovations

Assessment Method Evolution

Assessment Type	Traditional Approach	GenAI-Adapted Approach	Source
Written assignments	Essay completion	Dual-report method evaluating project results and prompt engineering proficiency	Jonathan W. Browning et al., 2024
Code evaluation	Manual code review	AI-aware assessment using CodeBERT for semantic similarity	Milena Nikolic et al., 2025
Clinical assessment	Traditional patient care evaluation	AI-generated care plans with human evaluation	M. Shen et al., 2025
Design projects	Final product assessment	GAI iterations combined with client feedback and tutor evaluations	Silvia Albano et al., 2025

Assessment Type	Traditional Approach	GenAI-Adapted Approach	Source
Group work	Team deliverables	Subgroups comparing AI-assisted vs. manual completion	N. Roy et al., 2025

Traditional assessment methods are increasingly viewed as ineffective in the GenAI era. Innovative approaches include requiring students to document why they accepted, modified, or rejected each AI suggestion, making reasoning explicit.

Learning Activity Modifications

Activity Type	Implementation	Outcome	Source
Reflective journaling	Students document AI usage and prompt strategies	Metacognitive insight development	N. Roy et al., 2025
Comparative analysis	AI-generated vs. human-created work comparison	Critical evaluation skills	Troy Schotter et al., 2025
Validation tasks	Peer-review of AI artifacts	Reduced passive reliance	Vahid Garousi et al., 2025
Real-world scenarios	Client-specific design projects using AI	Professional skill development	Silvia Albano et al., 2025
Ethical case studies	AI analysis combined with emotional reflection	Balanced technical and humanistic skills	M. Shen et al., 2025

The AI-Lab intervention produced large effect sizes in student comfort and openness to GenAI across conceptual, debugging, and homework problems. Students demonstrated more mindful and deliberate approaches to AI tool usage.

Skills and Competencies Framework

Technical GenAI Skills

Skill	Description	Training Approach	Source
Prompt engineering	Iterative query refinement for optimal AI outputs	Worked examples, guided exercises	Troy Schotter et al., 2025
Tool selection	Choosing appropriate GenAI tools for tasks	Benchmarking exercises across tool categories	N. Roy et al., 2025
Output validation	Verifying accuracy and completeness of AI responses	Validation tasks in assignments	Vahid Garousi et al., 2025

Skill	Description	Training Approach	Source
AI customization	Training GenAI for specialized applications	Student-built custom GPT tools	Yoram Reich et al., 2025

Students showed gradual improvements in using GenAI tools with statistically significant improvements across all GenAI literacy dimensions ($p < 0.0002$). One student even developed a custom GenAI tool by training ChatGPT-4o for specialized modeling tasks.

Critical Thinking and Evaluation

Competency	Evidence of Importance	Training Gap	Source
Hallucination detection	Students evaluated safety including accuracy and hallucinations	60-85% rated critical evaluation high	Multiple studies
Bias recognition	33-100% expressed concern about bias	Training interest: 38-88%	Remco Jongkind et al., 2025
Context evaluation	Students struggled with complex logical structuring even with AI	Over-reliance on AI-generated answers	Title Pending 6645
Source verification	Citation inadequacies identified as AI limitation	Students preferred content demonstrating originality	Cari Hyde-Vaamonde et al., 2025

Legal education students demonstrated sophisticated critical evaluation of AI-generated content, identifying limitations including superficial analysis, lack of argumentative coherence, and absence of nuanced understanding.

Ethical Awareness

Ethical Concern	Student Awareness Level	Educational Response	Source
Data privacy	46-76% top concern	Integration of ethics into curriculum	Multiple studies
Academic integrity	Significant concerns noted	Usage declarations required	Multiple studies
Environmental impact	13-28% concerned	Limited curricular attention	Remco Jongkind et al., 2025
Copyright/attribution	12-46% concerned	Formal instruction needed	Multiple studies
Algorithmic bias	Growing awareness	Bias mitigation training	Multiple studies

Ethical appraisal skills showed the lowest competency ratings among students (15-56%), indicating a significant gap that educational institutions must address.

Academic-Industry Gap Analysis

Identified Disconnects

Gap Area	Academic Practice	Industry Expectation	Source
GenAI competency valuation	Not emphasized in higher education	Increasingly valued	Anya Bardach et al., 2025
Practical application	Course-specific assignments	Technical documentation, data processing, decision-making	Andrea L. Irish et al., 2025
Critical analysis depth	GenAI assists with initial drafts	Detailed legal and critical analysis required	Armin Alimardani et al., 2024
Ethical implementation	Limited training focus	Responsible use essential	Multiple studies
Validation skills	Insufficient emphasis	Critical for quality assurance	Vahid Garousi et al., 2025

Faculty members noted GenAI's ability to assist with course-specific assignments but expressed concerns about over-reliance and ethical implications, while industry professionals emphasized efficiency gains through technical documentation and decision-making support.

Bridging Strategies

Strategy	Description	Evidence of Effectiveness	Source
Industry partnerships	Collaborative projects and internships	Alignment with workforce demands	Andrea L. Irish et al., 2025
Professional panels	Industry professionals discussing AI impact	Reassurance about career viability	Troy Schotter et al., 2025
Real-world scenarios	Client-facing design projects	Professional-quality outcomes	Silvia Albano et al., 2025
Career-aligned messaging	Communicating AI awareness importance in job interviews	Enhanced graduate preparedness	Vahid Garousi et al., 2025
Curriculum co-design	Faculty-industry advisory boards	Industry-aligned curricula	Andrea L. Irish et al., 2025

Only 54% of students believe employers will soon expect graduates to have GenAI knowledge and skills, indicating a perception gap that universities must address through better communication about industry expectations.

Implementation Evidence and Outcomes

Measured Learning Improvements

Intervention	Outcome Measure	Result	Source
AI-Lab framework	Comfort and openness to GenAI	Large effect sizes	Ethan Dickey et al., 2025
Systems engineering seminar	GenAI literacy dimensions	$p < 0.0002$ improvement	Yoram Reich et al., 2025
Nursing AI intervention	Self-rated competency	1-2 point increase on 5-point scale	Meghan Reading Turchioe et al., 2024
AI Immersion program	Technical proficiency, critical thinking	Improvements reported	Khinsa Fairuz Zahirah et al., 2025
SAGE framework	Movement beyond passive acceptance	84% of groups	Mahmoud Elkhodr et al., 2025
SPIRAL integration	Creative media and GenAI self-efficacy	Positive changes	Troy Schotter et al., 2025

The interventions demonstrate that structured pedagogical approaches can effectively enhance students' GenAI competencies while maintaining domain-specific skills.

Challenges and Limitations

Challenge	Description	Frequency	Source
Complex logical structuring	Students struggled even with AI assistance	Significant	Title Pending 6645
AI over-reliance	Tendency to accept AI answers without evaluation	Common	Multiple studies
Query framing difficulties	Students unable to frame queries for useful results	Essential skill gap	Jonathan W. Browning et al., 2024
System boundary identification	55% struggled with AI misclassifications	Majority	Mahmoud Elkhodr et al., 2025
Exception handling	55% overlooked missing exception handling	Majority	Mahmoud Elkhodr et al., 2025
Technical background gaps	Lower technical students required additional support	Variable	Khinsa Fairuz Zahirah et al., 2025

These challenges highlight that GenAI integration requires careful scaffolding and cannot simply replace traditional skill development.

Theoretical Frameworks and Future Models

Emerging Frameworks for GenAI Integration

Framework	Key Components	Application Context	Source
Causal model for responsible GenAI use	Professor actions, student factors, tool characteristics	Software engineering education	Vahid Garousi et al., 2025
SPIRAL integration	Domain skills first, then AI integration	Creative media education	Troy Schotter et al., 2025
Three-theme curriculum approach	Learning about, with, and aside from AI	Medical education	I. Shimizu et al., 2023
GenAI-CT framework	Ethical and pedagogical AI integration	STEM education	Title Pending 6645
Socio-technical strategy	Technical, organizational, personal perspectives	Nursing education	John Robert Bautista et al., 2025
Multiple perspectives theory	Technical, organizational, personal viewpoints	Computer engineering	Mostafa Mohamad et al., 2025

The three-theme approach from medical education provides a particularly comprehensive framework: learning about GenAI (literacy and ethics), learning with GenAI (efficiency and information support), and learning aside from GenAI (maintaining competencies AI cannot replace).

Generalizability of Findings

Claim	Supporting Evidence	Limitations	Source
Scalable curriculum innovation	AI Immersion program described as scalable	Single institution pilot	Khinsa Fairuz Zahirah et al., 2025
Transferable guidelines	Co-created legal education guidelines	Specific to legal context	Cari Hyde-Vaamonde et al., 2025
Cross-disciplinary AI-Lab	Framework recommended beyond CS	Primarily tested in CS	Ethan Dickey et al., 2025
International applicability	DigComp 2.2 framework basis	European context	Remco Jongkind et al., 2025
Multi-sector relevance	Insights for diverse educational settings	Survey-based evidence	Clara Rispler et al., 2025

Single-site implementation limits transferability of many findings, and cultural specificity may affect generalizability. However, common themes across diverse contexts suggest core principles may be broadly applicable.

Synthesis: Reconciling Heterogeneous Findings

Explaining Variation in GenAI Impact Perceptions

The literature reveals polarized perspectives on GenAI's impact, with engineering students showing a bimodal distribution of "P(doom)" estimates. This heterogeneity can be explained through several mechanisms:

Disciplinary Context: STEM students, particularly in computer science and engineering, show more frequent GenAI adoption and generally positive attitudes, while clinical and health sciences students demonstrate greater caution. This variation reflects differences in how directly GenAI tools apply to professional tasks.

Exposure and Experience: Students who have used GenAI in educational contexts show reduced fear of job displacement. The "demystification" effect suggests that hands-on experience enables more nuanced understanding of AI capabilities and limitations.

Institutional Support: Variation in institutional policies and infrastructure creates disparate student experiences. Most medical schools lack GenAI policies (93%) or training, while some computing programs have implemented comprehensive frameworks with measurable positive outcomes.

Reconciling Conflicting Evidence on Pedagogical Approaches

Studies show tension between embracing GenAI's productivity benefits and concerns about skill erosion. This apparent contradiction resolves when examining intervention timing and structure:

Sequencing matters: The SPIRAL approach demonstrates that developing domain skills before introducing AI tools leads to positive outcomes in both domain competency and AI literacy. Students who learn fundamentals first show better critical evaluation of AI outputs.

Scaffolding is essential: Unstructured GenAI use risks shallow learning, while structured interventions with explicit critical thinking components show improved outcomes.

Balance is achievable: Medical education's three-theme framework and nursing education's emphasis on balancing "high-tech" AI competencies with "low-tech" humanistic skills provide models for integration that preserve essential human competencies.

Implications for Different Career Categories

STEM Careers: The evidence strongly supports integrating GenAI as a core competency, with emphasis on validation skills and critical evaluation. Programs should focus on developing graduates who can work alongside AI rather than being replaced by it.

Trade Careers: Limited evidence suggests GenAI integration should focus on efficiency enhancement and problem-solving skills. The emphasis on performance expectancy and facilitating conditions indicates that practical, hands-on approaches may be most effective.

Arts Careers: The demystification effect observed in creative media and the successful integration in interior design suggest that arts programs can benefit from structured GenAI integration that enhances rather than replaces creative processes. The iterative nature of AI tools supports adaptive learning and real-time design refinement.

Recommendations by Institution Type

For **university programs** (STEM focus), evidence supports:

- Implementing validated frameworks like SPIRAL or AI-Lab
- Prioritizing assessment redesign over detection tools
- Developing longitudinal scaffolding of AI literacy across curriculum
- Establishing industry partnerships for real-world alignment

For **trade schools** (vocational focus), limited but emerging evidence suggests:

- Focusing on practical AI applications that enhance efficiency
- Addressing technological literacy gaps through structured training
- Implementing ethical guidelines around plagiarism and data privacy
- Building institutional support infrastructure before curriculum integration

For **arts schools** (creative focus), evidence indicates:

- Introducing AI after establishing foundational creative skills
- Using AI tools to support iterative design processes
- Addressing career anxieties through exposure and professional panels
- Emphasizing critical evaluation to distinguish human creativity from AI generation

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